Alpha Sentry

User Guide — from installation to every screen

A local-first research studio for WorldQuant BRAIN. It runs on your computer, uses your own BRAIN login and your own AI keys, and never submits an alpha to BRAIN on its own.



Alpha Sentry
the app



Autosentry
autonomous finder



Supersentry
SuperAlpha finder



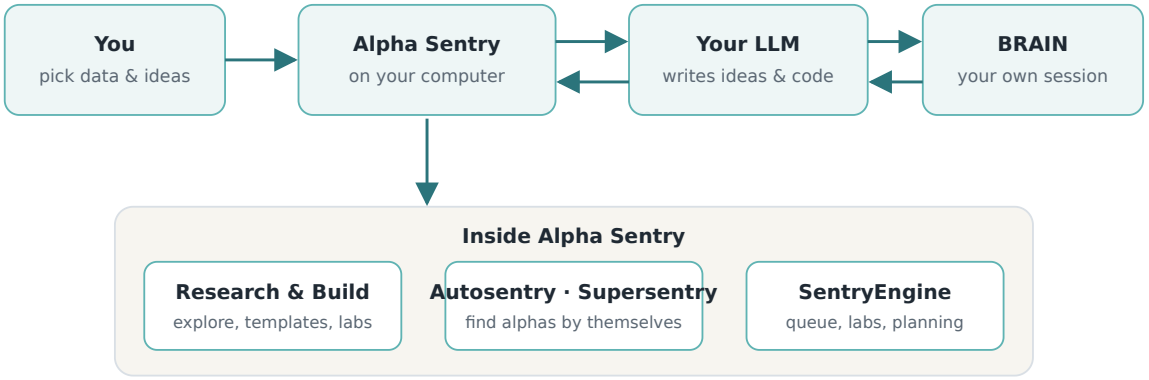
SentryEngine
batch engine &
labs

Nothing is submitted to BRAIN by the app · your keys and session stay on your computer · SentryEngine is off until you turn it on

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The big picture

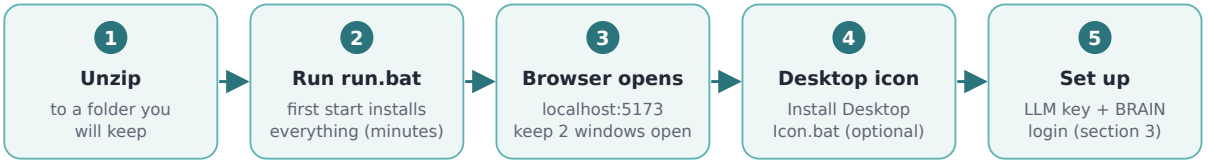


Everything runs on your computer. Only your LLM provider and BRAIN receive requests.

1. What you need

Need	Why
Windows 10/11 (macOS/Linux also work with `run.sh`)	The app runs on your computer
Internet	To reach BRAIN and your LLM provider
A WorldQuant BRAIN account	Alpha Sentry uses *your* session; nothing is stored on any server
At least one LLM API key (Claude, Gemini, OpenAI, DeepSeek, Groq, ...)	Ideas, templates and translations are written by an LLM
Python 3.11 or 3.12, and Node.js 18+	`run.bat` installs both if missing (Windows may ask permission: accept)

2. Installation



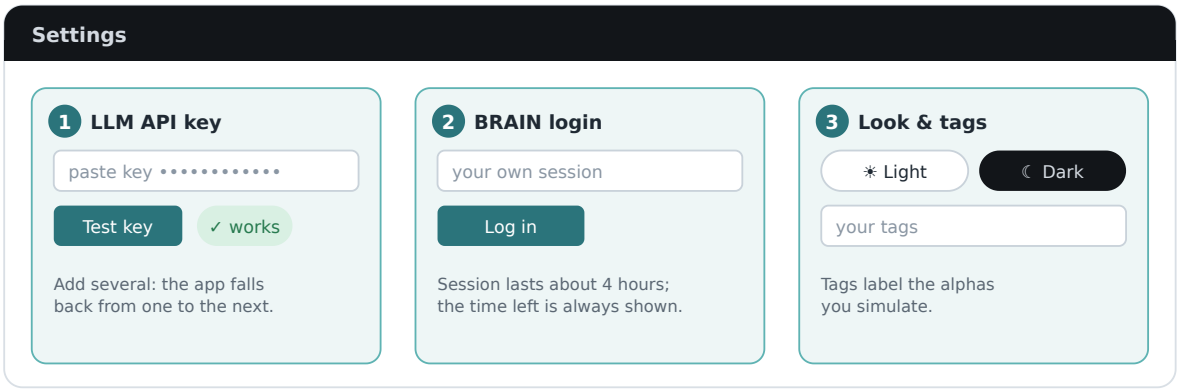
The five installation steps.

1. **Unzip** the Alpha Sentry folder somewhere you will keep it (for example `C:\AlphaSentry`). Do not run it from inside the zip. *Tip: a folder outside OneDrive avoids sync locks while packages install.*
2. **Double-click** `run.bat`. The first start takes a few minutes: it finds or installs Python and Node, creates a private Python environment, and installs the backend and frontend packages. Later starts take seconds.
3. Two windows open (backend and frontend) and your browser opens **http://localhost:5173**. Keep both windows open while you work; closing them stops the app.
4. **Optional desktop icon:** double-click `Install Desktop Icon.bat`. It creates **Alpha Sentry** and **Stop Alpha Sentry** on your Desktop. **Alpha Sentry** starts the app quietly in the background and opens it in its own window; closing the window leaves it (and any running Autosentry) going; **Stop Alpha Sentry** ends it. Logs are in `logs\`.
5. **Very long folder paths:** Windows breaks setup when the folder path is long (e.g. inside a WhatsApp transfers folder). Run.bat now copies the app to a short folder automatically. Best: unzip to `C:\AlphaSentry`. If setup still fails, send `logs\setup_error.txt` and `logs\pip.log`.
6. **Upgrading:** unzip the new version over the old folder; your data is kept. Delete any leftover `ACE Studio.vbs` and `Stop ACE Studio.vbs`, then run `Install Desktop Icon.bat` again.

If it does not start: read the last lines in `logs\`. "Network error — is the Alpha Sentry server running?" means the backend window is closed; start it again. `ECONNREFUSED 8766` means the backend is not running: `run.bat` starts both.

3. First-time setup (5 minutes)

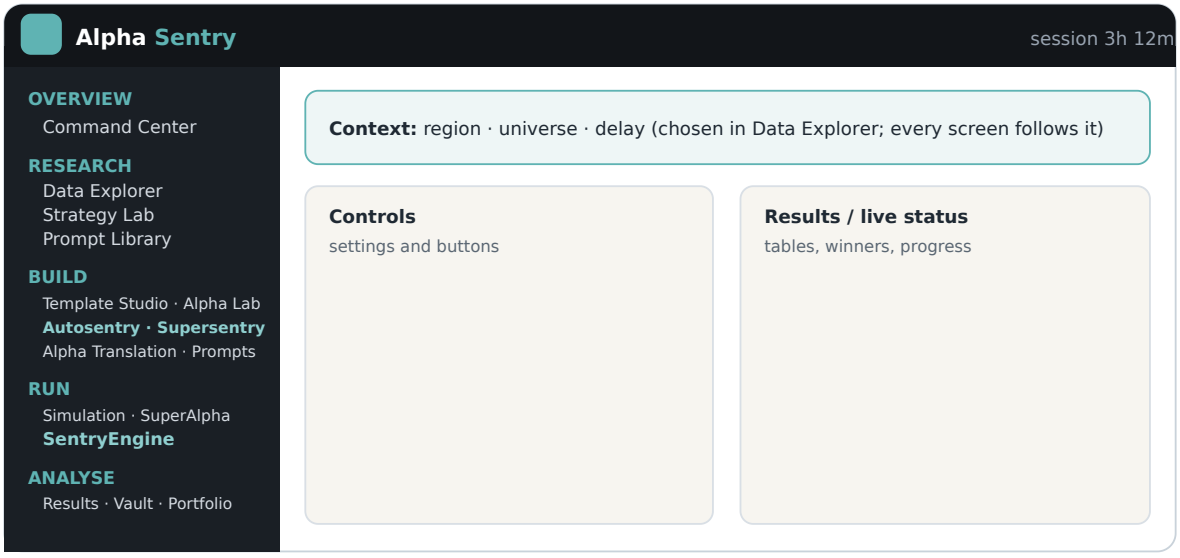
Open **Settings** (the gear icon in the top bar).



The three things to do in Settings (illustration; your screen's details may differ).

Optional: set up **Telegram** (section 9) to control runs and receive winners from your phone. The colours follow an eye-comfort scheme: soft charcoal in dark mode, warm cream in light mode, and a muted teal accent used only on small controls.

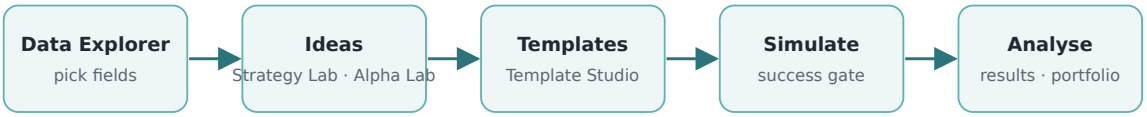
4. The layout



Sidebar on the left (screens grouped by job), top bar with the BRAIN session and Settings, work area on the right.

Name	What it is
Alpha Sentry	The whole application
Autosentry	The autonomous alpha finder
Supersentry	The same idea for SuperAlphas (selections and combos)
SentryEngine	The batch engine with labs, tasks and planning

5. The research workflow, step by step



You submit on the BRAIN site yourself, never from Alpha Sentry.

5.1 Command Center

Your dashboard: live success rate, how many alphas are submittable, diversity and the next best actions. Start here each day.

5.2 Data Explorer: choose what to work on

1. Pick the **context**: region, universe, delay.
2. **Fetch** datasets and fields from BRAIN (always live; MATRIX and VECTOR fields are marked).
3. Narrow with the family-prefix filter and presets, then **select** fields (select-all and random are available).

Your selection is what Autosentry and the generators build around.

5.3 Strategy Lab and Prompt Library

Strategy Lab: per-field and per-category hypotheses and suggested operators you can send straight to Alpha Lab. **Prompt Library**: save the dataset-focused prompts you like and export them.

5.4 Template Studio

Write templates with `{field}` and `{field2}` placeholders. It checks syntax as you type, completes operators with Tab, and can ask the LLM for variants.

5.5 Alpha Lab

Turns a hypothesis (typed, or from a research paper) into validated expressions, repairs invalid ones, steers away from ideas you already have, and lets you push them to Simulation.

5.6 Simulation: test expressions

1. Paste or send expressions in.
2. Choose universes, neutralizations, tags and concurrency.
3. Run. Results are judged against the success gate (section 12).

5.7 SuperAlpha and BRAIN Prompts

SuperAlpha builds selections of at least 10 components, counts them and simulates combos in its own workspace. **BRAIN Prompts** builds the prompt text for BRAIN's LLM-prompt submissions and keeps the form fields.

6. Autosentry and Supersentry

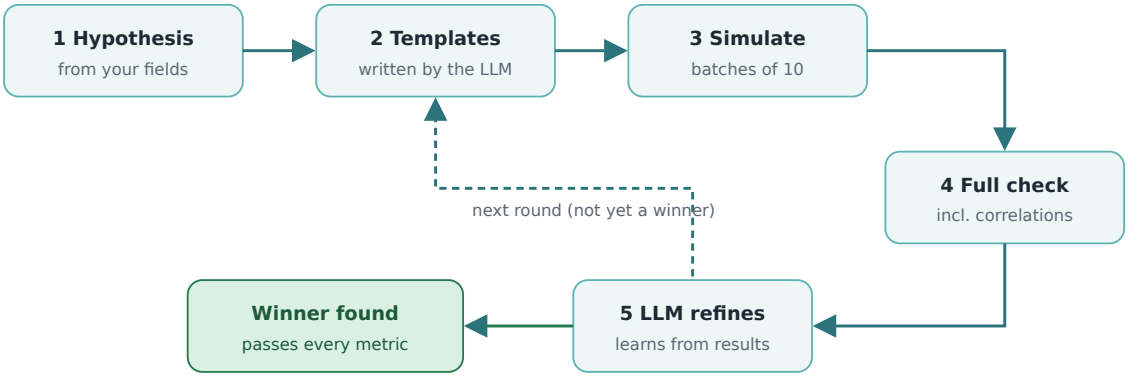
Markets: Region, Delay and Universe are dropdowns at the top of Autosentry, Supersentry and every SentryEngine screen, filled from the data stored on this computer (plus BRAIN's list when loaded).

Winners are always kept and announced: every alpha Autosentry finds is tagged in BRAIN, saved to the Alpha Vault, added to the library and sent to your Telegram chat (not silent) with Sharpe, fitness, turnover, correlations and the expression. A step that fails never stops the others and the log says which. **Prod rescue limit:** an alpha that fails only on production correlation is fixed-up unless the correlation is above 0.74 (setting "Skip prod rescue above").

Pace: each round tests one batch of 10 alphas ("Batches per round" = 1; raise it to run several side by side). Production-correlation checks run one at a time with a wait before each ("Prod check delay", 60 seconds by default; 0 = no wait, up to 4 at once).



Autosentry runs by itself until an alpha passes **everything**, and keeps going until you press **Stop**.



The Autosentry loop.

To use it

1. Select fields in Data Explorer first.
2. Open **Autosentry** and press **Start Autosentry**.
3. Watch the phase, cycle and round, the current hypothesis, how many alphas were simulated and the winners.
4. Press **Stop** to end it after the current step.
5. **Reverse and boost rule:** an alpha is reversed only when its Sharpe is below −1, and every alpha with Sharpe above 1 is boosted (one threshold; a booster's own limit is ignored). These follow-ups ride in the next standard batch of 10 and never run as separate simulations.
6. **Boosters:** with none in your Library, built-in starters are used (neutralize, decay, rank, zscore, liquid days only). Each alpha gets its 3 most promising boosters, ranked by past gain and by the check it failed, plus one exploratory; boosters that keep hurting are retired. Promising alphas also get window nudges.
7. **Test many ideas:** each idea gets one batch of 10; if its best |Sharpe| is below 0.5 it is dropped and the next idea starts. Promising ideas continue for up to 4 rounds.
8. **Clone and swap:** switch the method at the top of the Autosentry panel. Old alphas (submitted and/or vault) keep their shape and get similar datafields from the Data Catalog swapped in; a slider runs from "same dataset" to "other datasets". Batches of 10, then the full check incl. prod correlation; winners go to the vault with their lineage.
9. **Reading the log:** every batch logs what was queued (or why nothing was); a run summary appears when it stops.

The Library

Your personal library of alphas and boosters, plus the lessons learned. Each item has **region ticks**:

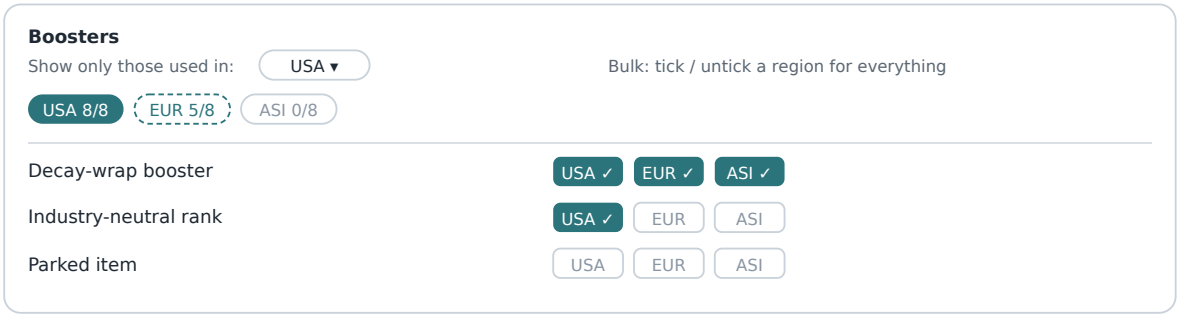
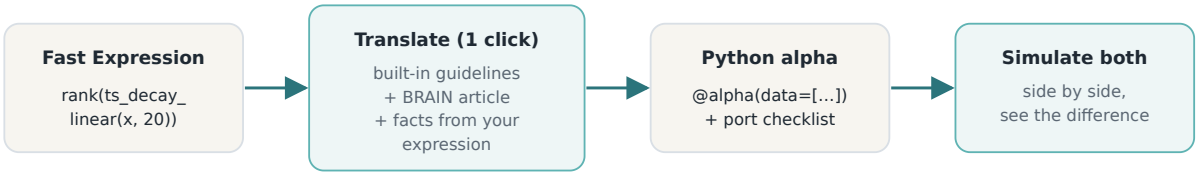


Illustration of the region ticks (item names are examples).

- All ticked (default) = used in every region; untick regions to restrict; untick all to **park** the item.
- The bar above the list ticks or unticks a region for everything at once; a dashed pill means "some".
- When you add an item, choose its regions on the spot.
- A new installation starts with an **empty** library and no lessons, so yours grows from your own work.

> **Supersentry** is the same idea for SuperAlphas. It shares BRAIN's simulation slots with Autosentry, so run one at a time.

7. Alpha Translation: Fast Expression to Python



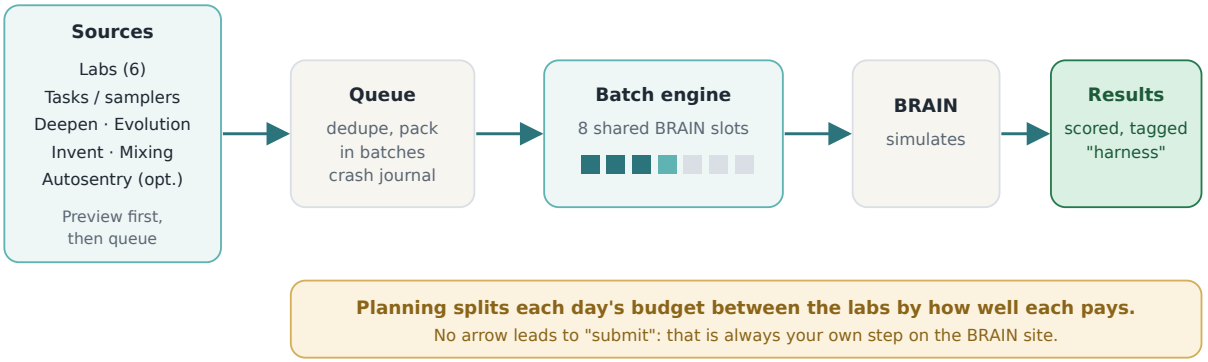
If BRAIN reports an error: "Send error back to the LLM to fix".

1. Paste a **Fast Expression**.
2. Check the simulation settings (they default to your context; keep them the same as the original so the comparison is fair).
3. Press **Translate to Python**. It works in **one click**: the translation guidelines are built in (your checklist plus the full BRAIN article, included word for word).
4. The Python code appears (you can edit it) with a **port checklist**: field types, operators the guideline does not cover, and the cold-start days.
5. **Simulate** the Python alpha and the original side by side.
6. If BRAIN reports an error, press **Send error back to the LLM to fix**.

> **Guideline:** "View / edit guideline" shows the built-in text. Save your own to override it; **Restore built-in** brings the original back. Tag the translated alpha with the original alpha's ID.

8. SentryEngine: engine, labs, tasks, planning

An optional power layer. Everything in it is **off by default** and it **never submits** anything to BRAIN: it only simulates, scores and queues. Results land in Results & Analytics with the tag `harness`.



How work flows through SentryEngine.

8.1 Control tab

- **Feature flags:** turn on "SentryEngine + labs" first. Other switches enable the returns vault, LLM labs, planning and the options below.
- **Batch engine:** Start / Stop, the queue (queued, running, done, scored) and the 8 shared slots.
- **Reset engine:** a clean slate in one press. Cancels everything queued or running in SentryEngine, stops its search tasks, clears its region focus, crash journal and learned slot limit, frees its slots and starts a fresh engine. Results, vault, library and Autosentry's running batches are not touched; nothing is submitted.
- **Follow Autosentry's region (on by default):** SentryEngine always works on the region chosen in Autosentry. When you change it, and at every engine tick after, queued simulations for any other region are dropped immediately and their search tasks paused. Running batches finish normally. Untick the box in Queue by region to park instead of drop.
- **Queue by region:** the queue is shared by every region and one BRAIN batch holds only one region. Pick your region and press **Work on this region**: other regions are parked (kept), or dropped with the tick box, and their search tasks paused. **Drop** (or **Drop everything**) removes a region's queued simulations and abandons the ones already sent to BRAIN: the engine stops waiting, frees their slots and ignores their results, and a dropped simulation never comes back. A batch BRAIN never finishes is given up after 90 minutes so its slot is freed.
- **Yield by lab:** how many passing alphas each lab produced recently.
- **Autosentry speed:** with "Autosentry via engine" on, each round can run more batches side by side (up to 8, versus Alpha Sentry's own 3). It lowers itself when BRAIN is busy and lets a very slow batch finish on its own so a round is not held back.
- **Shipped data:** every version carries the dataset and field catalogue, and the operator reference. It contains no one's alphas, results, vault or library. On start, rows your database lacks are added (never changed or removed), after a one-time backup. It does **not** include any Library data.

8.2 Tasks, Deepen, Evolution, Invent, Mixing

Tab	What it does	Use it when
Tasks	One template with `$variables`; a sampler (random, grid, Sobol, NSGA-II/III, TPE, CMA-ES) picks combinations and learns. **Power Pool** mode keeps alphas within 8 operators and 3 fields. Pause, resume or stop any task.	You have an idea with open choices (windows, fields, decay)
Deepen	Builds a template exploring around one winner	You want to know if a winner is real or a lucky setting
Evolution	Genetic search from your winners under your limits	You want relatives of winners, not just tuned copies
Invent	Your LLM proposes templates; each is checked before it can be queued	You want fresh shapes
Mixing	Finds alphas that combine well from stored returns and correlations	You have several passing alphas
Planning	Splits each day's budget across labs by yield, explains itself in words	You run several labs

8.3 The six labs

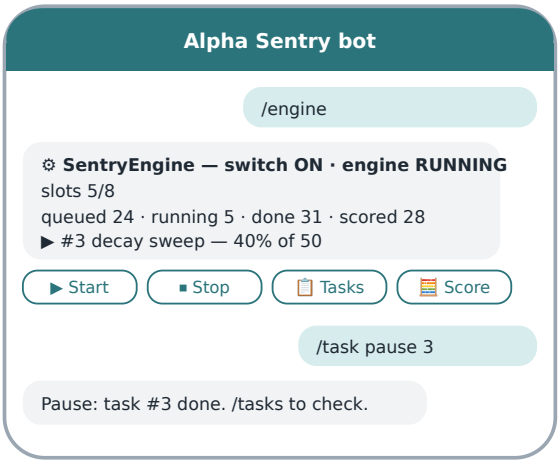
Lab	What it does	Use it when
Relocate	Runs formulas that work in one market in the others (skips markets lacking a needed field)	A winner exists in one market
Repair	One targeted fix per near-miss, chosen from the check it failed	Many alphas almost pass
Harden	Nudges a winner's windows and wrappers to see if it holds	Before trusting a winner
Harvest	Sensible patterns over the best fields of each dataset	Cheap, broad exploration
Diversify	Uses only fields from datasets none of your alphas use	Your alphas are too alike
Pair	Cross-dataset field pairs in a few relational shapes	You want combinations of signals

Every lab has **Preview** (nothing is queued) and **Run** (queues simulations in the engine). The **Guide** tab inside SentryEngine explains each feature again in the app.

> **Typical first session:** Control → turn on "SentryEngine + labs" → Labs → pick one lab → **Preview** → **Run** → Control → **Start** the engine → watch Results.

9. Telegram remote control

Set it up once in the Autosentry panel (Telegram setup): create a bot with BotFather, paste its token, use **Find my chat ID** and send a test. Only your own private chat is obeyed; everyone else is ignored.



An example chat (illustrative numbers).

Command	What it does
`/menu`	Buttons for everything below
`/status`	What is running, round, winners, session time left, and the SentryEngine line
`/winners`	Latest winners
`/start regular` · `/start super`	Start Autosentry or Supersentry with their last settings
`/stop` · `/restart`	Stop, or stop and start again
`/switch regular` · `/switch super`	Stop the running one and start the other
`/settings` · `/set …`	View or change settings (`/neut FAST,SLOW` , `/region DEU`)
`/refresh` · `/session`	Renew the BRAIN session now · time left
`/engine`	SentryEngine status: switch, running, slots, queue, running tasks
`/engine on` · `/engine off`	The "SentryEngine + labs" switch
`/engine start` · `/engine stop`	Start or stop the batch engine (stop lets running batches finish)
`/engine score`	Score finished simulations into Results now
`/tasks` · `/task pause|resume|stop <id>`	List tasks · control one

When your BRAIN session expires, Autosentry pauses and Telegram sends the biometrics link. Nothing is ever submitted to BRAIN from Telegram.

10. Analysing results

- **Results & Analytics:** every simulation with per-metric verdicts; filter, sort, export CSV, see which operators perform.
- **Alpha Vault:** the alphas you keep, with their returns.
- **Correlation & Portfolio:** pairwise PnL correlation and the largest set you can submit together.
- **Operator Atlas** and **R & U Atlas:** operators with your own usage overlaid; regions and universes with market notes and cross-region transfer.
- **Data Catalog:** a one-time download of datasets and datafields by region, delay and category (shared by Autosentry, Clone and swap and SentryEngine), with browse and CSV / Excel export.
- **Osmosis Planner:** reads every submitted alpha, proposes the 100,000-point allocation for each qualifying scope, and writes your edited points to BRAIN on request.

8.10 Slots, Results and prod checks

Simulation slots (Control) shows eight tiles: who holds each slot (engine task, Autosentry or Supersentry), region, number of simulations and time elapsed; amber after 20 minutes; click a busy engine tile to abandon it.

Results → Missing results lists every simulation that never reached Results with one plain reason, counts per reason, a search box, and **Score now** / **Requeue**.

Results → Alpha stats lists all alphas with their stats and source; red rows are alphas not stored in Results. **Check prod** checks one alpha; **Check selected** goes one by one with a 60 s wait before each and a Stop button. Prod above 0.70 is red; above 0.74 it is marked too high. Nothing is simulated or submitted.

Clone and swap ignores Python alphas.

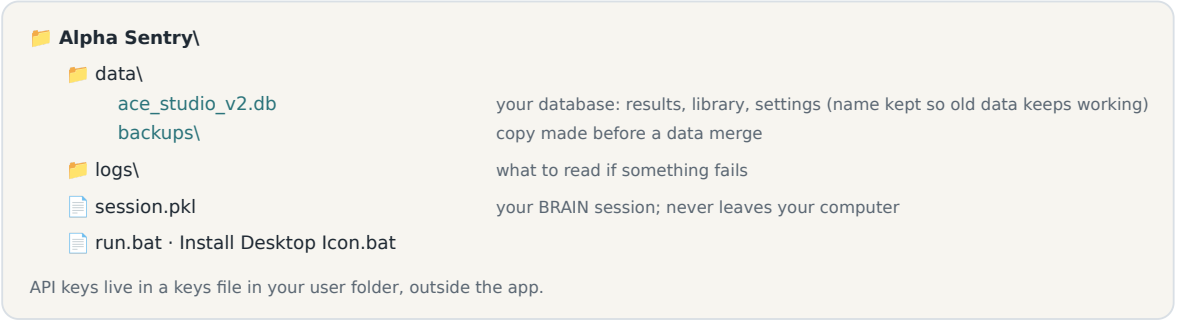
Osmosis Planner

Reading is read-only. It only ever writes the Osmosis points on your alphas, and only when you press **Implement on BRAIN**. It never submits an alpha.

1. Press **Read my alphas from BRAIN**: it lists submitted alphas and downloads their daily PnL.
2. Each alpha is screened: big Sharpe loss under investability constraints (unless unique on its dataset), turnover above 20% unless robust, correlation above 0.70 with a better alpha.
3. A scope **qualifies** with at least 10 passing alphas and gets exactly 100,000 points, spread by inverse volatility with caps per alpha, dataset, category, neutralization, operator and high-turnover share.
4. BRAIN needs 3 allocated scopes. Only allocated scopes count toward your daily rank, so weak scopes are left out unless needed to reach 3.
5. Scopes that do not qualify show how many alphas they are short.
6. **Edit and implement:** the points are suggestions you can change; each scope shows its total live (exactly 100,000 over 10+ alphas). **Implement on BRAIN** confirms, writes the points, reads BRAIN back and shows each scope's total. **Undo last** restores the previous points. **Download plan (CSV)** is there if you prefer to enter points yourself.

It is an estimate from your own PnL and cannot reproduce your rank against other consultants. Allocations lock every Sunday 23:59 EST.

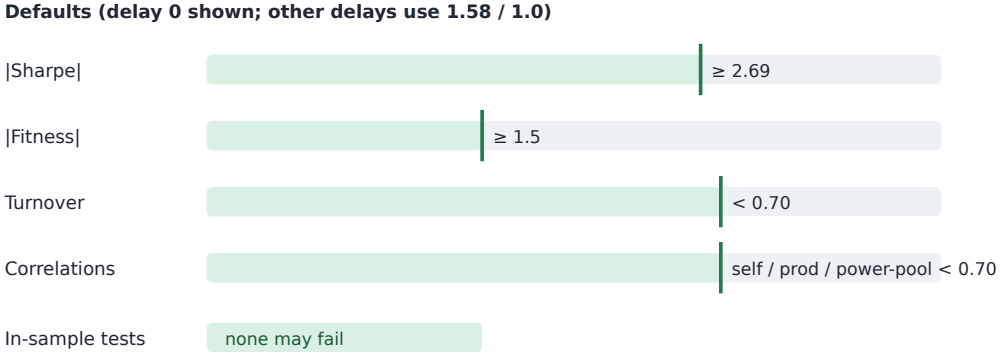
11. Your data, backups and privacy



- Everything stays on your computer. Only your LLM provider and BRAIN receive requests.
- **Back up:** copy the `data` folder while the app is stopped.
- **Start fresh:** stop the app and move the `data` folder away.

12. The success gate

An alpha is a verified success only when **every** check passes, in absolute value.

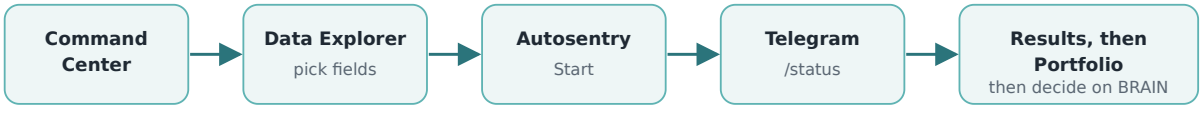


Green = passing side of each threshold. Thresholds are editable.

13. Troubleshooting

Problem	What to do
App will not open	Make sure the backend window is open; run `run.bat` again
Install fails part-way	Move the folder outside OneDrive (for example `C:\AlphaSentry`), install Python 3.12 and Node LTS yourself if the automatic install failed, then run `run.bat` again; read `logs\`
"Session expired" or missing	Settings → log in to BRAIN again, or `/refresh` in Telegram
A key test fails	Re-paste the key, check the provider's credit, try another provider
Autosentry rounds are slow	SentryEngine → Control → turn on "Autosentry via engine" and review its speed
SentryEngine buttons do nothing	Turn on "SentryEngine + labs" in Control first
Telegram does not answer	Check the token and chat ID in Telegram setup; only your registered private chat is answered
SentryEngine is stuck and nothing helps	Control → Reset engine (clean slate; your results are kept)
SentryEngine seems stuck after changing region	Control → Queue by region → pick the region and press "Work on this region" (or "Drop" for the old one)
Clone and swap stops with "Nothing to swap"	The log says why: source datafields not in the Data Catalog for this region/delay, every swap already simulated, or no similar field. It widens the search once before stopping
429 / "slots busy" from BRAIN	You are at BRAIN's limit; the app slows down by itself; do not run Autosentry and Supersentry together
Desktop icon does nothing	Run `Install Desktop Icon.bat` again from the current folder

14. Quick reference



A daily routine

> **Rules that never change:** nothing is submitted to BRAIN by the app; your keys and session stay on your computer; SentryEngine is off until you turn it on.